

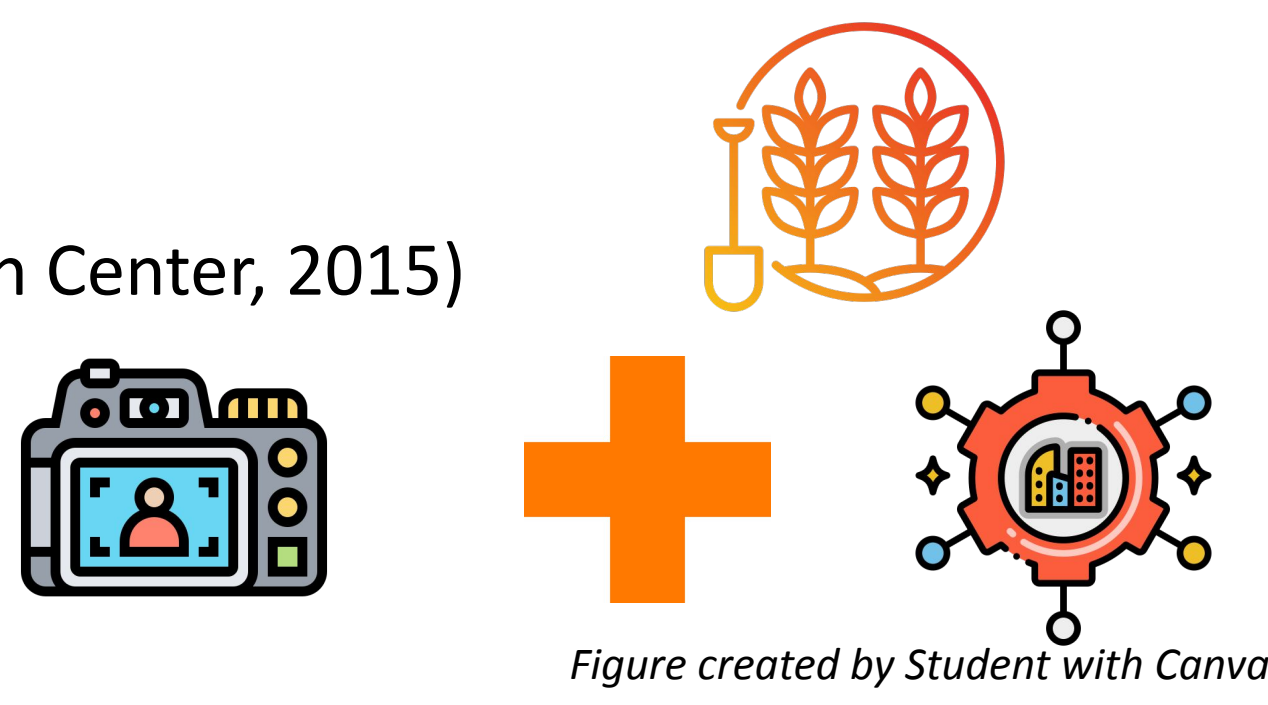
Securing DJI Drones: A Real-Time Wi-Fi Intrusion Detection and Response System

Joseph Park

Introduction

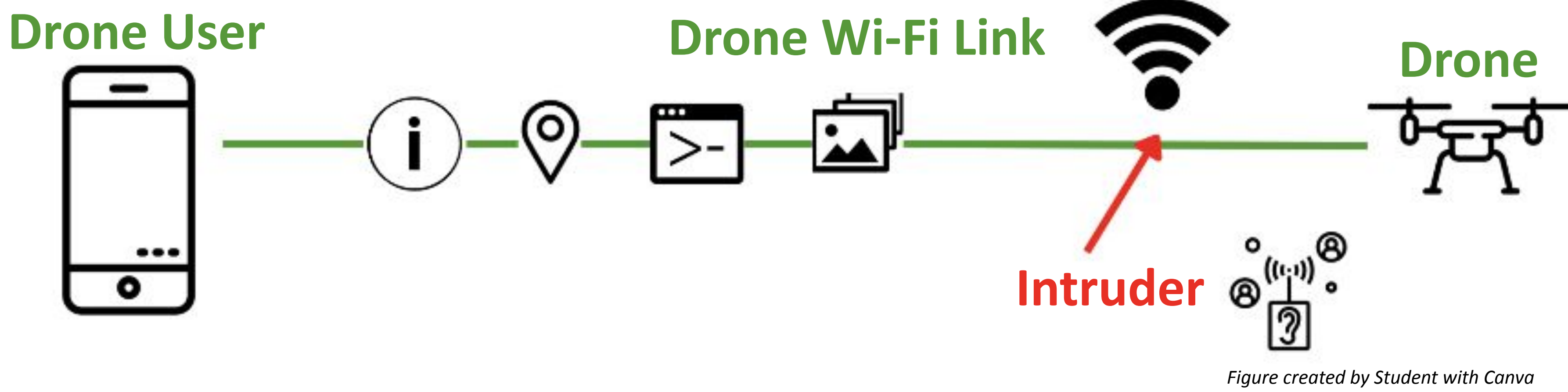
Drone Prevalence

- Drone usage **surging** in USA and beyond
 - Est. 1 in 12 Americans** own a drone (Pew Research Center, 2015)
- Drone applications enormous and extensive
 - Agriculture, critical infrastructure, etc.**



Drone Security & the Wi-Fi Vector

Insecure Wi-Fi networks used for communication



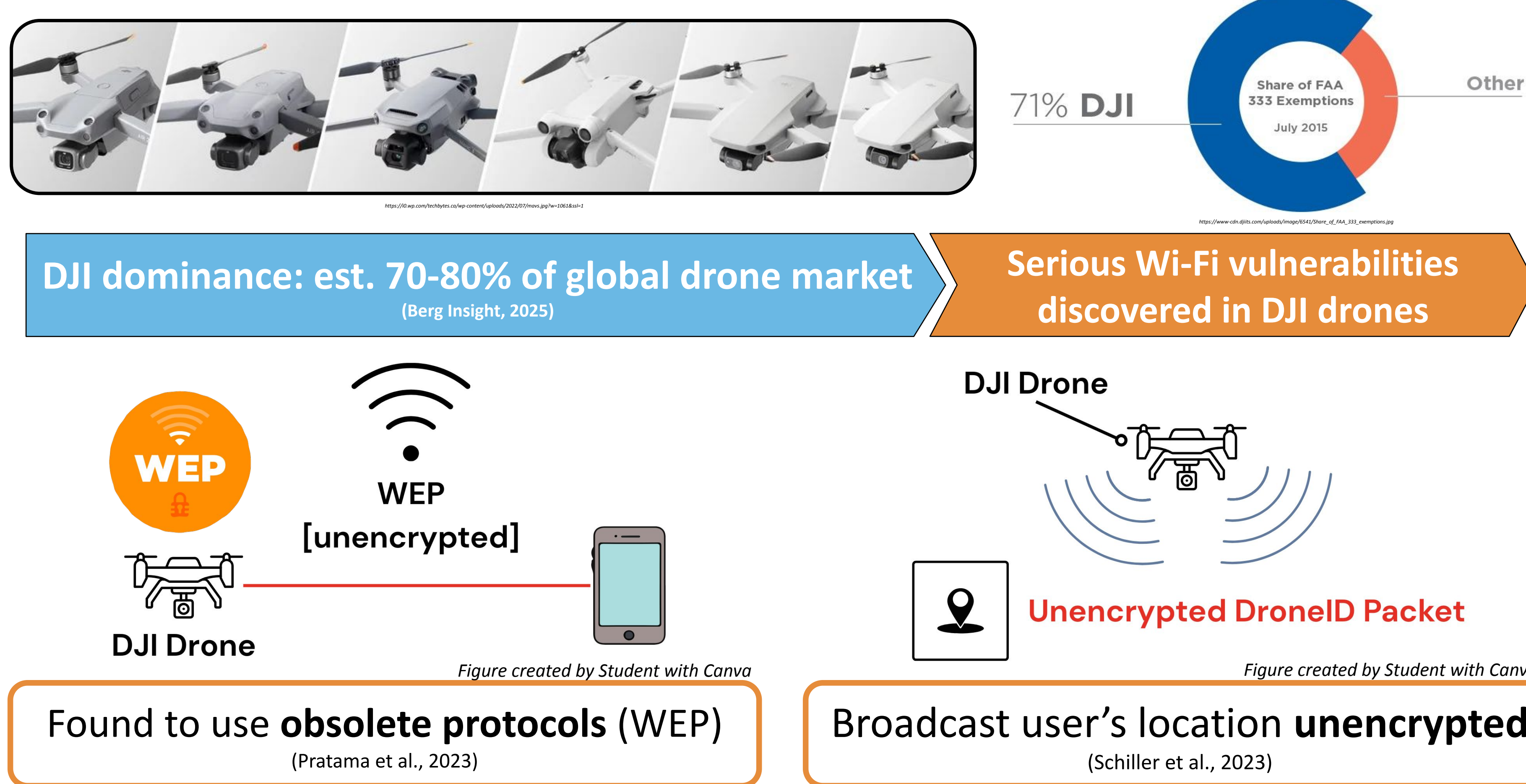
Deprecated encryption (e.g. WEP)

Insecure credentials/passwords

Lack of protective mechanisms (IDS)

Review of Literature

DJI Drones: Documented Wi-Fi Vulnerabilities

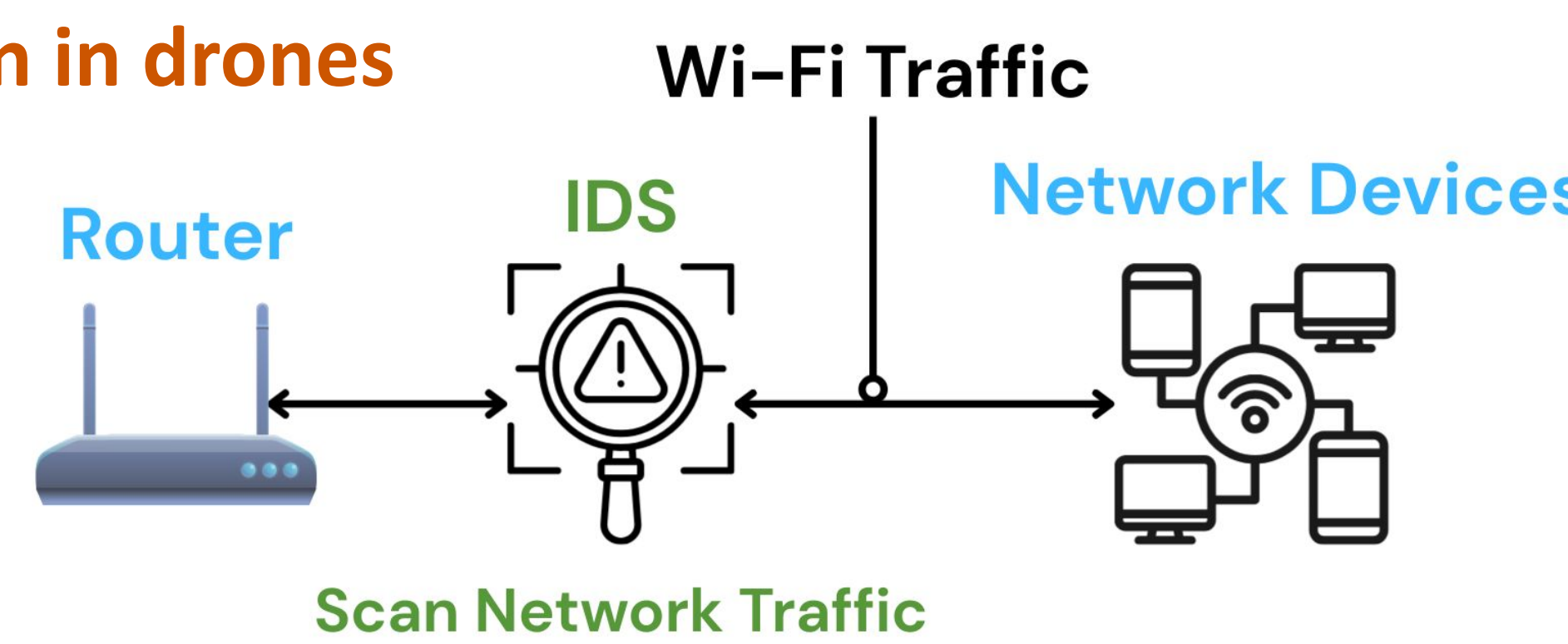


Intrusion Detection Systems (IDS)

Intrusion Detection Systems (IDS) typically used to protect enterprise networks

Promising but limited implementation in drones

- Cyber-Physical IDS for UAVs (Hassler et al., 2023)
- Adversarial, GAN-based GPS IDS (Asif et al., 2023)



Problem // Research Gap // Goals

Problem

Consumer drones (especially DJI models) lack real-time, live-tested Wi-Fi defensive systems

Research Gap

Real-time, *Wi-Fi-focused* defensive systems for drones are underdeveloped/rare:
Limited by simulations and/or offline testing

Goals

- [Phase I] - Create a threat model of a DJI drone
- [Phase II] - Create a real-time IDS that detects and mitigates Wi-Fi attacks

Methodology

Materials

[Provided by Mentor]

- DJI Neo Drone
- Alfa Wi-Fi USB Adapter

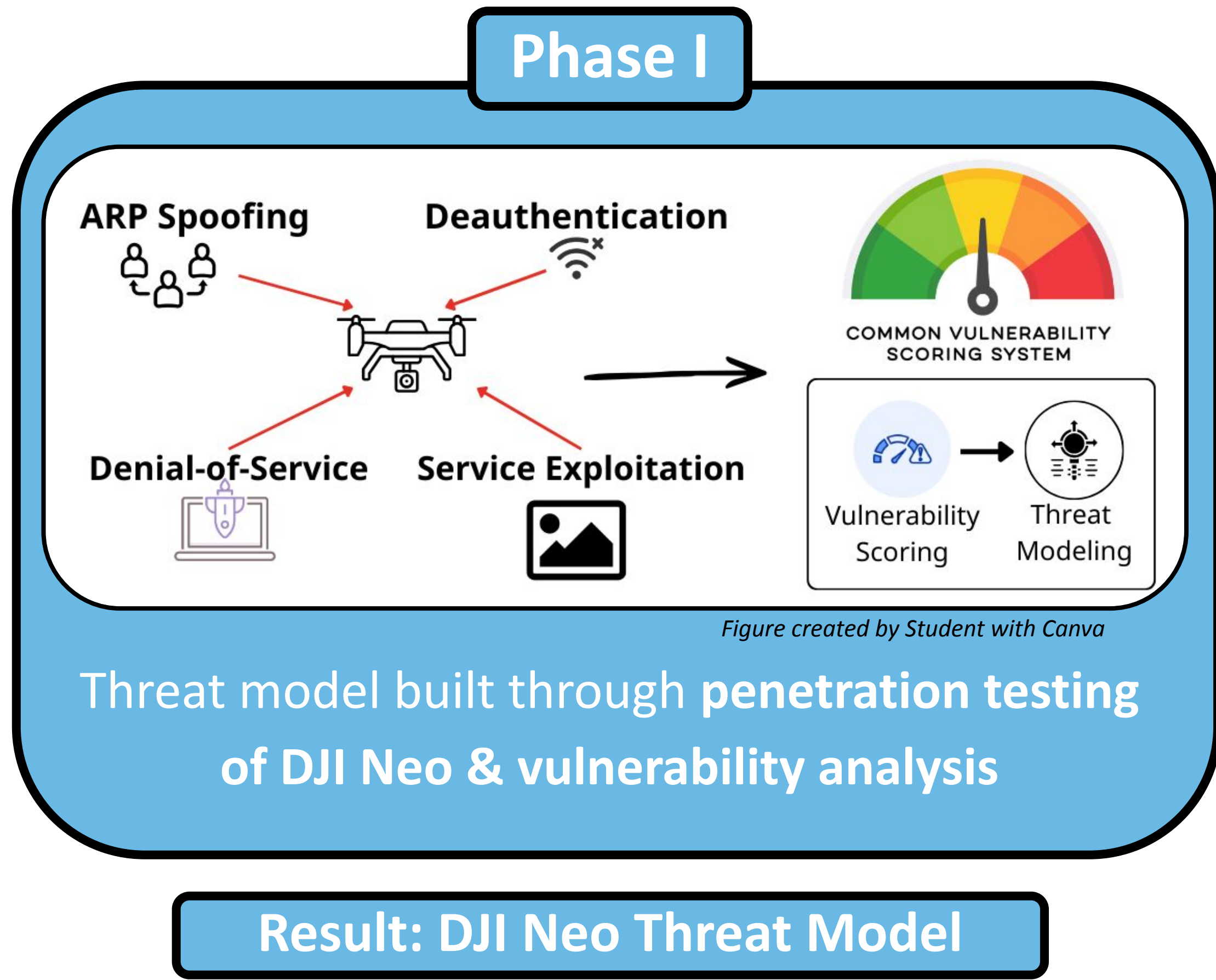
Experimentation

[Conducted by Student]

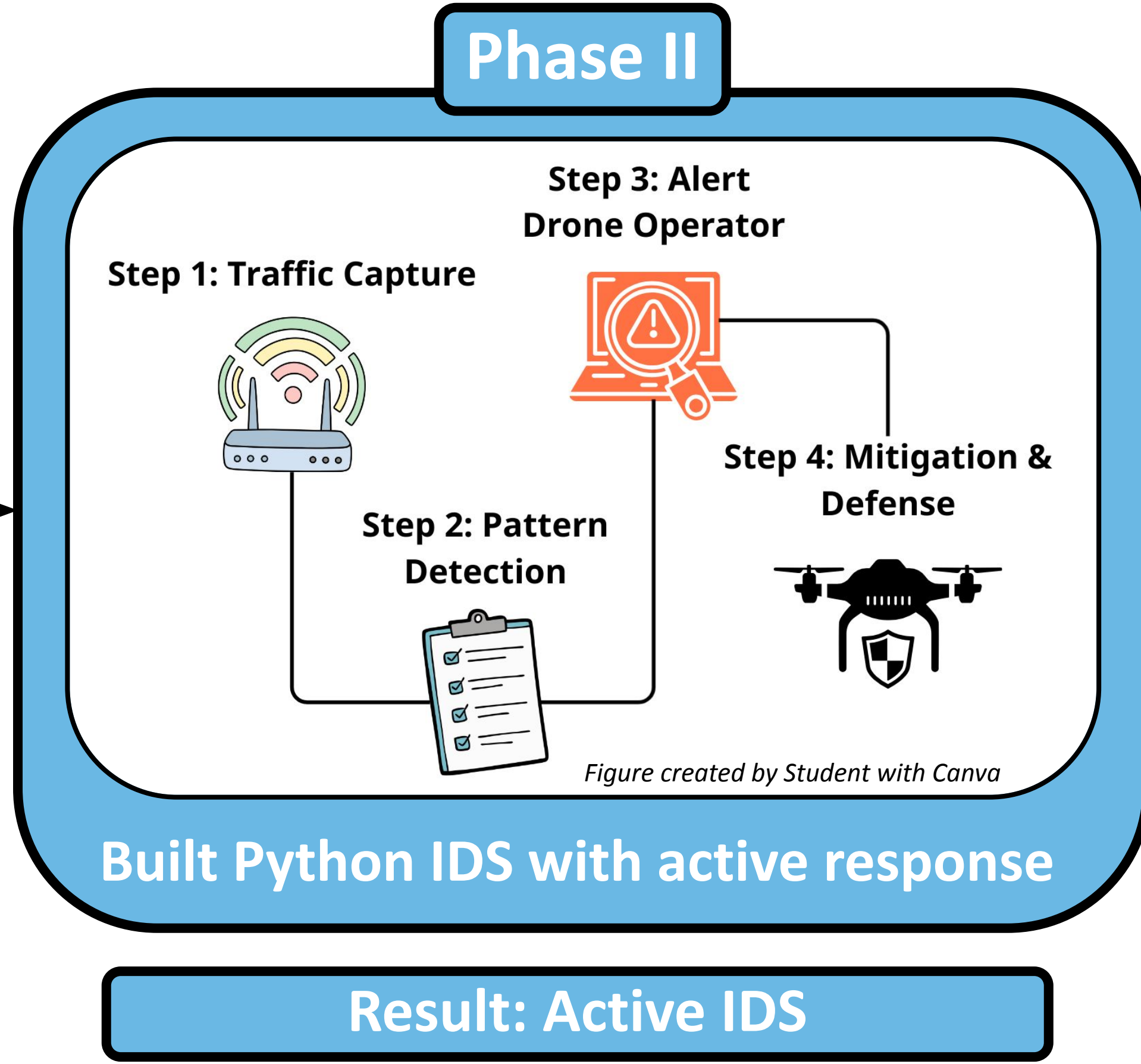
- [Phase I]
 - DJI Neo Security Assessment
 - Threat Modeling
- [Phase II]
 - IDS Design & Development
 - Live Testing



Figure created by Student with Canva

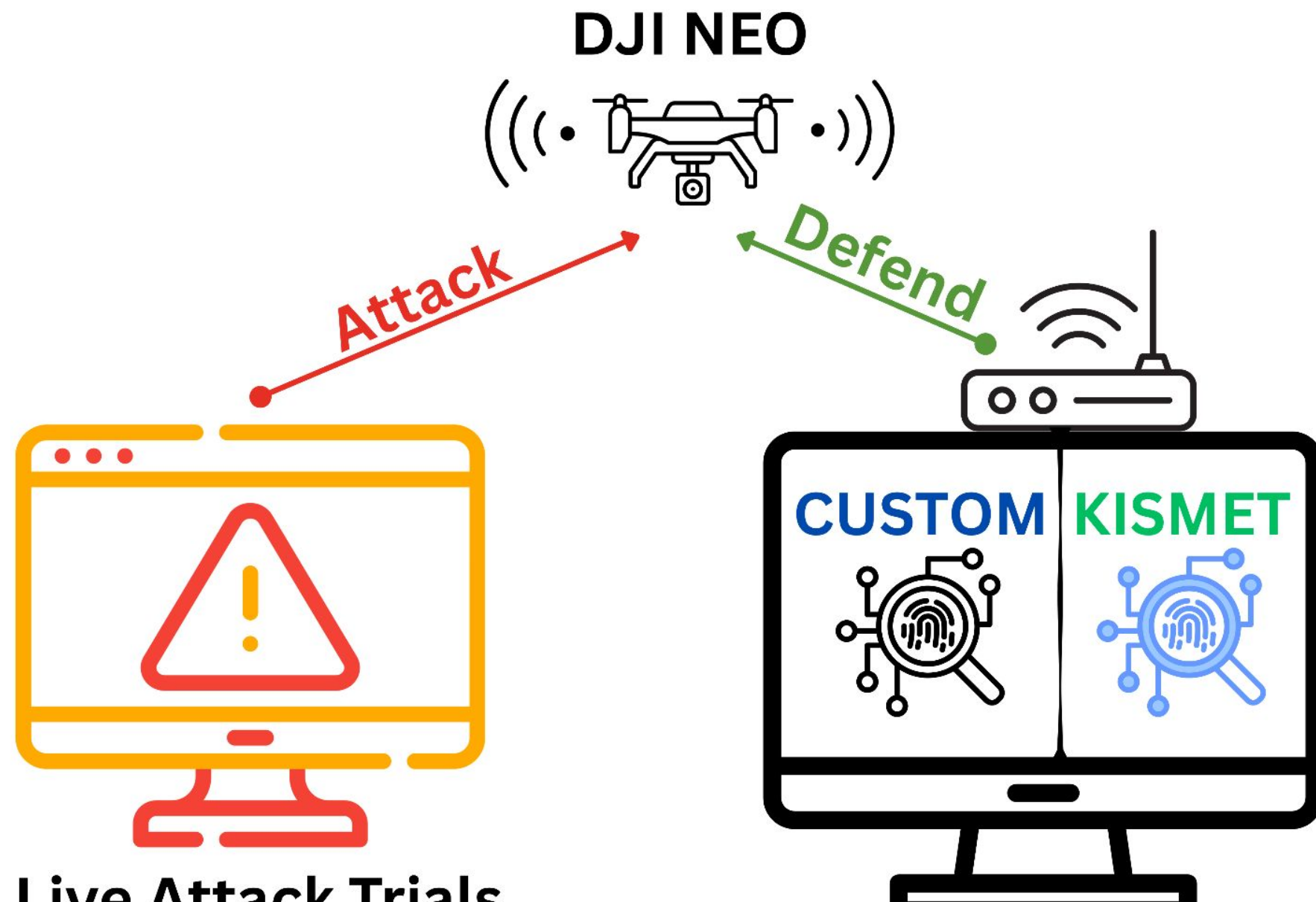


Result: DJI Neo Threat Model



Result: Active IDS

Live Evaluation vs. Kismet (Industry IDS)



Evaluate IDS in live attack scenarios

Run Kismet in parallel for comparison

Results

Phase I: Vulnerability Testing

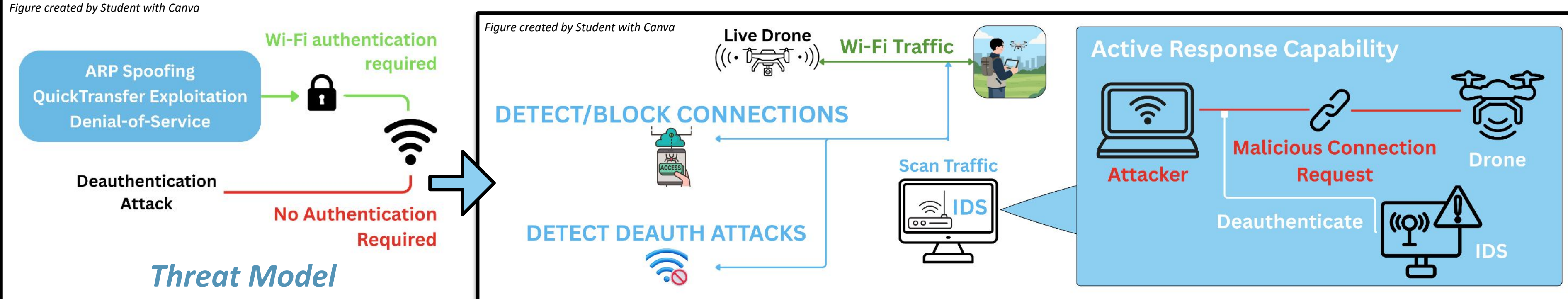
Vulnerability	CVSS v4.0 Score	CVSS v4.0 Severity
Deauthentication Attack	7.1	HIGH
Service Exploitation	5.9	MEDIUM
ARP Spoofing	4.6	MEDIUM
Denial-of-Service	3.5	LOW

Figure created by student researcher using Google Sheets

DJI FLY screen capture - produced by student

Wireshark screen capture - unauthorized file download - produced by student

Phase II: IDS Development



Phase II: Live Evaluation

Accuracy & Detection				
Attack Type	Trials	IDS Accuracy	Kismet Accuracy	Active Response Accuracy
Unauthorized Connection	5	100.00%	100.00%	100.00%
Deauthentication	5	100.00%	100.00%	N/A
Combined Attack	3	100.00%	100.00%	100.00%
Overall	13	100.00%	100.00%	N/A

Figure created by student researcher using Google Sheets

Detection Speed			
Attack Type	Trials	Mean IDS Detection Latency (seconds)	Mean Kismet Detection Latency (seconds)
Unauthorized Connection	5	2.214	1.200
Deauthentication	5	2.370	1.800
Combined Attack	3	5.385	3.000
Overall	13	3.006	1.846

Figure created by student researcher using Google Sheets

Resource Consumption			
	IDS CPU Usage	Kismet CPU Usage	IDS Memory Usage
Mean	21.29%	32.45%	5.90%
Median	26.00%	22.00%	6.06%
95th Percentile	34.00%	82.34%	6.19%
Maximum	46.00%	124.00%	6.25%

Figure created by student researcher using Google Sheets

Discussion

Phase I: Vulnerability Patterns

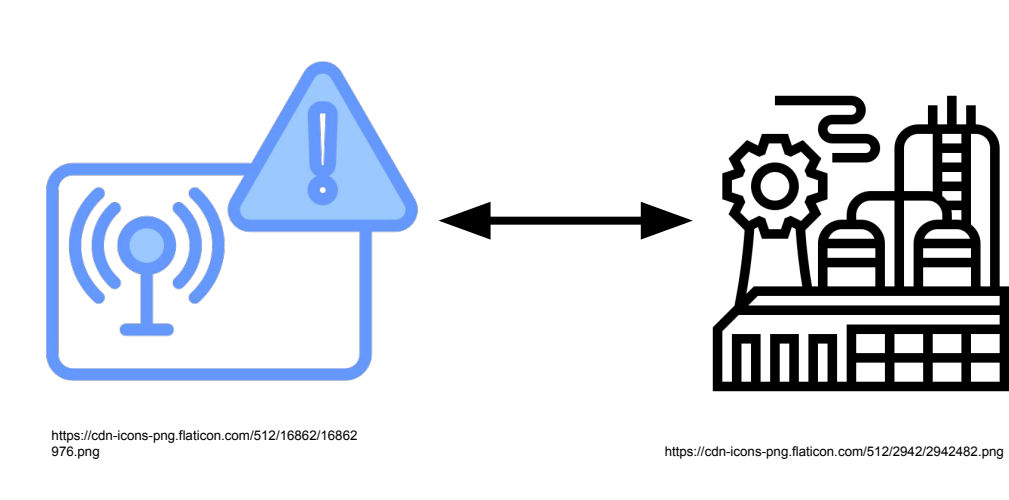
Patterns: Weak Encryption, Authentication/Security Mechanisms

Consistent with previous studies; systematic, structural flaws

Phase II: Study IDS vs. Kismet (Industry IDS)

Study IDS

- Accuracy: 100%
- Stable resource usage
- Lower CPU utilization



Kismet (Industry IDS)

- Accuracy: 100%
- Faster detection speed (by ~1.2s)
- Less memory required

Study IDS Differentiator: ACTIVE RESPONSE CAPABILITY

Automatically detects and blocks unauthorized connections to the drone's Wi-Fi network

Limitations & Future Research

- 1 drone model, only indoor conditions tested
- Narrow attack focus (no jamming, GPS spoofing)
- Single-drone system; fleets not evaluated
- Test with multiple DJI models in outdoor conditions
- Expand threat coverage: jamming, RF attacks
- Integrate detection features for multiple drones

Applications

Emergency Response

- Protect emergency response operations from cyberattacks

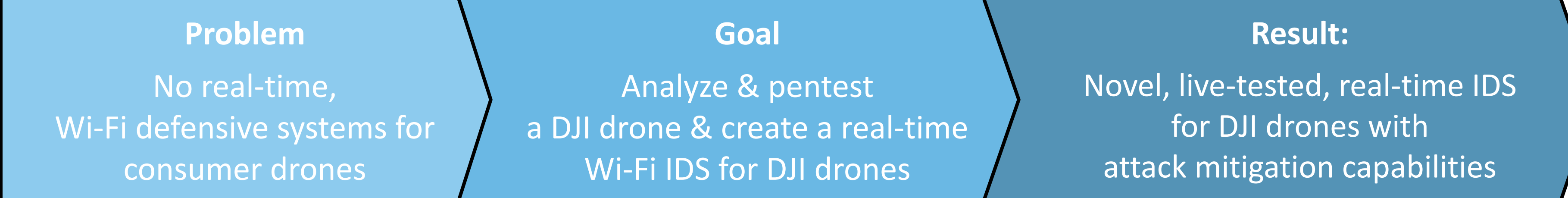
Personal Users

- Sensitive information (photos, GPS data), expensive equipment

Critical Infrastructure

- Power infrastructure, stations, turbines
- Roads, pipelines

Conclusion



Novelty

- New vulnerabilities discovered in DJI Neo model
- First live-tested Wi-Fi IDS for DJI drones in literature
- Automatic, *active* mitigation; superior to passive industry systems

References

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